MIL-A-45016C(AL)
6 February 1985
SUPERSEDING
MIL-A-45016B(WC)
22 October 1968

MILITARY SPECIFICATION

ALIGNMENT SERVICE UNIT MOTOR VEHICLE WITH FRAME STRAIGHTENING AND ALIGNMENT CAPABILITIES

This specification is approved for use by the Armament Research and Development Center (Tool and Equipment Engineering Directorate; Standardization and Specification Engineering Division), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a frame straightening unit for detecting and correcting misalignments and distortions of frames, axles and axle housings. The frame straightening unit shall also have the capability of performing wheel alignment.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense of Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-C-843 - Cushioning Material, Cellulosic
PPP-T-60 - Tape, Packaging, Waterproof

MILITARY

MIL-C-104 - Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-C-3774 - Crates, Wood: Open, 12,000 and 16,000 Pound Capacity

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods

2.1.2 Other Government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B30.1 - Jacks

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.2.1 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete frame straightening unit for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The frame straightening unit shall be new and shall be for permanent installation, the design embodying a rack and a complete line of equipment for checking and straightening vehicle frames and axles, and aligning wheels. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, and replacement.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the manufacturing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The manufacturer is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.

3.4 Construction. All parts used in fabrication shall be new. The frame straightening unit shall be complete so that when connected to the specified power source, it can be used for any operation specified herein. There shall be no process used for reclaiming any parts.

3.4.1 Castings and forgings. All castings and forgings shall be free from defects affecting the structural integrity and load carrying capacity, scale and mismatching.

3.4.2 Welding, brazing and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered part. These operations, however, shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All parts subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently deformed.

3.4.4 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards which can be guarded without substantial interference to operation of the frame straightening units. All guards shall be removable to provide easy access to guarded parts. Except as otherwise specified by the procuring activity, the frame straightening units shall comply with the standards promulgated under OSHA 2206 applicable to the alignment service units themselves. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1).

3.4.5 Threads. All threaded parts shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.4.6 Gage dials. All indicating dials shall be marked in the English system. The markings shall be permanent and shall be in plain, bold faced characters that can be easily read.

3.4.7 Interchangeability. All replaceable parts shall be designed and manufactured to such standards as shall permit replacement and adjustment without the need for modification of the parts or the frame straightening unit.

3.4.8 Surfaces. All castings, forgings, molded parts, stampings, and welded parts shall be clean and free of sand, dirt, fins, sprues, scale, flux, and other extraneous materials. External working surfaces shall be smoothed and free of burrs, with all edges rounded or beveled.

3.5 Performance and product characteristics.

3.5.1 Functional requirements. The frame straightening unit shall incorporate the facilities and capabilities to indicate misalignment, indicate the degree of correction required, and correct:

a. Areas of frame misalignment due to sag, sway, mash, buckle, twist, and diamond, without removal of the frame or axle from the vehicle.

b. Axle (I-beam and tubular) caster, camber and steering axis inclination by straightening. Rear and drive axle housing camber and toe by straightening without heating.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.

c. Vehicle overall suspension alignment, including caster, camber, steering axis inclination, turning radius, and toe in.

3.5.2 Capacity. The frame straightening unit shall be capable of performing the functions specified herein on Military vehicles within a weight range of no less than 2,000 to 40,000 pounds net weight involving a tread width range of no less than 48 to 94 inches and a wheelbase range of 80 to no less than 249 inches.

3.5.3 Overload. Ramps and vehicle supporting areas of the frame straightening unit shall be capable of withstanding weights of at least 50,000 pounds without bending, failing or damage.

3.5.4 Accuracy. All gages and measuring devices furnished with the frame straightening unit shall be accurate within 3 percent of the total scale range unless otherwise stated herein.

3.6 Details of components. The frame straightening unit shall consist of, but shall not be limited to, the following principal parts.

3.6.1 Straightening rack. The straightening rack shall be of the elevated type for floor mounted installation. The rack shall incorporate ramps, runways and provisions for mounting turning radius gages. The rack shall be constructed for frame, axle and body straightening and for use with gages and equipment specified in 3.6.4 through 3.6.6.2. The rack shall permit the use of the gages, wheel alignment equipment and frame and axle straightening attachments from the end, side or underneath the rack. The rack shall be fabricated to withstand the vehicle loads (see 3.5.2 and 3.5.3) and the pressure loads exerted when performing the operations specified in 3.5.1. Means shall be provided for anchoring the rack to the floor to provide a permanent rigid installation.

3.6.1.1 Straightening rack runways. Two runways shall be provided with a length no less than 25 feet and no less than 15 inches wide. The height of the driving surfaces of the runways from the floor shall not be less than 24 inches nor greater than 32 inches high. The distance between runways shall be adjustable to accommodate the tread widths specified in 3.5.2. The runways shall have means to prevent movement when loading or unloading a vehicle and means to lock the runways when used in the straightening operations.

3.6.1.2 Approach ramps. Two ramps shall be furnished, each not less than 15 inches wide. When attached to the unit, the ramp slope shall be no more than 3 inches of rise in 12 inches of horizontal run measured from the floor to the runway surface. The ramps shall be detachable and equipped with a positive locating device that shall securely hold the ramps in place to prevent ramp movement when the vehicles are being loaded and unloaded. The ramps, when attached to the unit, shall be capable of supporting the vehicle weights specified in 3.5.2 and 3.5.3.

3.6.1.3 Turning radius gage base. A "pan" or indentation shall be incorporated into the front of each runway to allow for mounting the turning radius gages. The indention shall be deep enough so that the surface of the turning radius gage on which the wheel sets is flush with the surface of the runway. The portion of the base below the level of the runway surface shall be wider than the runway to allow a person to stand when making alignment corrections, and shall cover the web of I-beams or H-beams cut to make the indention. There shall be no sharp edges or exposed beam webs on the turning radius gage base.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.

3.6.2 Frame straightening unit. The rack shall have a primary capability of frame straightening when used with the attachments specified in 3.6.4, but it shall also have a secondary capability of wheel and axle alignment when used with the attachments specified in 3.6.5 thru 3.6.6.2. The frame straightening unit shall comply with the requirements of 3.5.1 and 3.5.2. The unit shall also incorporate a method for maintaining beam/traveling member position and pressure to allow the jacks to be retracted and repositioned to gain added travel.

3.6.3 Hydraulic pressure jacks. The force required for the axle and frame straightening operations shall be applied by means of remotely operated type hydraulic rams. The hydraulic rams shall conform to ANSI B30.1. Each ram shall be equipped with a safety relief valve to prevent actuation when loads greater than the rated capacity are encountered. The use of an electric-hydraulic pump to operate the rams is acceptable. The minimum ram requirements for axle and frame straightening shall include four-20 ton, one-30 ton, and two-60 ton rams. Additional hydraulic units to accomplish other requirements specified in 3.5 shall be provided in accordance with the manufacturer's commercial practice for the model being furnished. The hydraulic equipment shall be used with the axle and frame straightening attachments to correct axle and frame misalignment as specified in 3.5.1.

3.6.4 Frame straightening attachments. Frame straightening attachments shall be provided for use with the straightening rack specified in paragraph 3.6.1 when correcting frame misalignments. The number of attachments shall be no less than the following:

2 - beams for diamond corrections and as a base support for straightening sagged and twisted frames.
4 - transverse beams used for holding and locking carriage uprights and as the traveling members when making frame corrections.
4 - uprights with detachable bars and heads used for transmitting force to the frame and for holding or raising frames. The uprights shall be capable of being locked into place.
6 - jack locks and jack supports for use on the beams when applying pressure with the jacks.
2 - jack stands and jack holders for use in raising and holding the jacks above the runways.
3 - towers or abutments used for holding and applying pressure when making "high pulls".
4 - pull bars 8 feet long threaded on one end and a clevis on the other used for holding and pulling frame and body members and washers and nuts for them.
4 - brackets for use with the pull bars. The brackets shall allow attachment of the pull bars at an angle. The brackets shall be used with the beams for pulling frame members.
2 - hooks and loops for holding down frames.
1 - wrench for removing wrinkles from frame flanges.
At least 2 hanger plates and handles for each beam.
Other parts as needed for use with the above.

3.6.5 Axle straightening and alignment adjusting attachments. Axle straightening and alignment adjusting attachments shall be provided for use with the straightening rack when adjusting caster and camber of I-beam and tubular axles and axle housings and in straightening the axles and axle housings. The number of axle straightening/alignment attachments shall be no less than the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.

1 - bridge beam for straightening axles/aligning wheels. The bridge beam is not required if the design of the straightening rack allows the use of a transverse beam or the straightening rack frame itself.
3 - heavy duty alloy steel chains of not less than 5/8 inch diameter wire with a self-locking ring or hook on one end and not less than 6 feet long.
2 - cables not less than 3/4 inch diameter with a ring attached to each end and at least 8 feet long for use between dual wheels when correcting axle housings.
2 - twisting tools or hooks for use when adjusting caster or twisted I-beam axles.
4 - arms or connectors used with yokes or clevises for adjusting camber. Two connectors for light duty and two for heavy duty use shall be furnished.
4 - yokes or clevises for use in conjunction with the connectors in adjusting camber on I-beam and tubular axles and axle housings. Pins for the above.
4 - steel wheel chocks, to block wheels.
3 - V-blocks for performing camber corrections.
4 - slotted plates for taking slack out of the connectors.
4 - yokes or clevises, utilized for correcting camber, caster, and axle correction on "I" beam and tubular heavy duty axles.

In addition to the above items, the alignment service unit shall include other necessary equipment required for performing the operations specified herein.

3.6.6 Gages.

3.6.6.1 Wheel alignment gages. The frame straightening unit shall incorporate not less than the following gages for checking the wheel alignment characteristics:

a. A set of turning radius gages shall indicate in degrees the turning angle of a wheel when the other wheel is turned 20 degrees in and 20 degrees out. The indicator scales shall not be less than 25-0-25 degrees with graduations of not more than 1 degree and numerical markings every 5 degrees. The gage shall have an accuracy of +1/2 degree. The gage shall be adequate to check all vehicles with weights and tread widths as specified in 3.5.2 and be designed with full floating action enabling the turning plates to rotate and simultaneously move laterally/horizontally. The turning radius gages shall have pins to lock the gages in a non-moveable position at 0 degrees. The gages shall be mounted in the base provided on the runways. Provisions shall be provided to protect all bearings and moving parts from dirt, sand, mud, water or any other contamination.

b. A tire scriber shall be provided for scribing the centerline of the tire tread when checking toe.

c. A wheel toe gage shall be provided to measure at hub height the distance in inches the tread width varies from the front to the back of the tires. The distance is measured between lines scribed around the centerline of the tires and can be measured on both front and rear wheels. The gage shall be capable of measuring toe on wheels of 10 to 24 inches in diameter and tread widths specified in 3.5.2. The pointers shall be adjustable so that the toe is measured at hub (center) height. The gage shall provide a measuring range of not less than 1/2 inch toe-out to zero to 1/2 inch toe-in and shall be graduated in not more than 1/16 inch increments with an accuracy of $\pm$ 1/32 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.

d. A caster gage shall measure in degrees the angle a line through the center of the ball joints, kingpin or struts varies from plumb vertical when the vehicle is viewed from the side. The scale shall have a range of not less than minus 9 degrees (the line tilted toward the front of the vehicle) to 0 to plus 9 degrees (line tilted toward rear of vehicle) with graduations of not more than 1/2 degree and an accuracy of ± 1/16 degree. The gage shall be marked in not more than 1 degree increments.

e. A steering axis inclination gage shall indicate in degrees the angle between the plumb vertical tire centerplane and a line through the center of the ball joints, kingpin or strut as viewed from the front or rear of the vehicle at zero degrees camber. The gage shall have scales no less than minus 9 degrees to 0 to plus 9 degrees graduated in 1/2 degree increments marked at 1 degree intervals and accurate to within at least ± 1/16 degree. The caster and Steering Axis Inclination (SAI) gages may be combined.

f. A camber gage shall indicate in degrees the angle the wheel tilts at the top from the vertical as viewed from the front or rear of the vehicle when the wheels are in a straight ahead position. The gage shall have a scale range of no less than plus 5 degrees (outward tilt of the wheel) to zero to minus 5 degrees (inward tilt of the wheel) with graduations of not more than 1/4 degree and an accuracy of ± 1/16 degree. The scale shall be marked at no more than 1 degree intervals. The caster, camber and steering axis inclination gages may be combined into a single unit.

3.6.6.2 Axle and frame alignment gages. The frame straightening unit shall include gages to measure and check the axle and frame alignment characteristics as stated below:

a. No less than 5 frame centering gages shall be provided for checking the frame to accurately locate areas of sway, sag and twist. The gages shall have an accuracy of at least ± 1/32 inch.

b. Alignment gages and a steering wheel holder shall be provided for determining the correct relationship between the front wheels and the complete steering system and for centering the steering. The gages shall be for tread widths of 48 to 94 inches (see 3.5.2) and wheel rim diameters of 10 to 24 inches and shall be designed to fasten to the wheel, hub or spindle. The gages shall have an accuracy of at least ± 1/32 inch. Templates or master gages shall be furnished for checking and calibrating the gages.

c. A wheel tracking gage shall be provided to test the squareness and parallelism of the front and rear wheels. The gage shall be adjustable for use on vehicles with wheel bases from 80 to no less than 249 inches. The tracking gage shall be mechanical, optical, electronic or a combination of these. If mechanical, the gage shall have no less than 4 adjustable contact points. The gage shall have an accuracy of at least ± 1/32 inch and shall mount on the wheel, hub or spindle. The alignment gages and tracking gages may be combined.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T02:14Z
Check the source to verify that this is the current version before use.

3.6.6.3 Gage finish. All gages shall be inherently corrosion resistant or shall be finished to resist corrosion.

3.7 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall be clean and be free of all foreign matter detrimental to painting and at least one coat of primer and one coat of enamel, or equivalent, applied.

3.8 Product identification. The alignment service unit shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.9 Workmanship. Standards of workmanship shall be such as to assure equipment of a neat general appearance which complies with the requirements in section 3. All components, subassemblies, parts, and the final assembly shall comply with tests in section 4.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).

b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production items (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.3 and all tests under 4.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of "a" through "b" below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

a. Product examination (see 4.3).

b. Accuracy test (see 4.4.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.

4.3 Product examination. Each frame straightening unit shall be examined to determine conformance with the requirements of Section 3, using the following checklist as a guide.

EXAMINATION CHECKLIST

CHARACTERISTICS	REQUIREMENT PARAGRAPH
Design 1/	3.2
Materials 1/	3.3
Reclaimed materials	3.3.1
Construction 1/	3.4
Castings and forgings 1/	3.4.1
Welding, brazing and soldering 1/	3.4.2
Fastening devices	3.4.3
Safety and health requirements 1/	3.4.4
Threads 1/	3.4.5
Gage dials	3.4.6
Interchangeability 1/	3.4.7
Surfaces	3.4.8
Straightening rack	3.6.1
Straightening rack runways	3.6.1.1
Approach ramps	3.6.1.2
Turning radius gage base	3.6.1.3
Frame straightening unit 1/	3.6.2
Hydraulic pressure jacks 1/	3.6.3
Frame straightening attachments	3.6.4
Axle straightening and alignment adjusting attachments	3.6.5
Wheel alignment gages	3.6.6.1
Axles and frame alignment gages	3.6.6.2
Gage finish 1/	3.6.6.3
Painting 1/	3.7
Identification of product	3.8
Workmanship	3.9

1/ A certificate of conformance may be provided to satisfy the examination requirement where more than a visual or dimensional verification is necessary.

4.4 Test methods.

4.4.1 Accuracy test. Each gage and measuring device furnished with the frame straightening unit shall be tested for accuracy by comparison with standard measuring equipment accurate to within 25 percent or less of the accuracy limits specified for the gage or measuring device (see 3.5.4 and 3.6.6 through 3.6.6.3).

4.4.2 Functional test. Utilizing a 1/4 ton, 4 by 4, M38lA1 utility truck and a 5 ton, 6 by 6, M54 cargo truck, the various components of the frame straightening unit shall be manipulated and operated to the extent necessary to verify the unit's ability to successfully perform the following operations on each vehicle:

a. The location and correction of areas of frame misalignment or distortion (see 3.5.1).

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T02:14Z
Check the source to verify that this is the currentversion before use.

b. Location and correction of front and rear axles and axle housing misalignments or distortions (see 3.5.1).

c. Location and correction of areas of misalignment or maladjustment in front end suspension system (see 3.5.1).

4.4.3 Load test. Using the approach ramps, a loaded or unloaded vehicle with a total weight of at least 50,000 pounds shall be driven onto the frame straightening rack, parked for five minutes, then backed off the rack. Any evidence of permanent deformation or damage to the rack or ramps shall be cause for rejection of the frame straightening unit (see 3.5.2 and 3.5.3).

4.5 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of section 5.

5 PACKAGING

5.1 Disassembly. Each frame straightening unit shall be disassembled in accordance with the requirements of MIL-STD-1186.

5.2 Preservation. Preservation shall be level A or industrial as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. The frame straightening unit shall be cleaned in accordance with process C-1 and dried in accordance with procedure D-1 or D-4 of MIL-P-116.

5.2.1.2 Preservative application. Immediately after cleaning and drying, a coat of type P-2 preservative shall be applied to all critical ferrous metal surfaces of the frame straightening unit and accessories. All remaining ferrous metal surfaces, subject to corrosion, shall be preserved with an application of type P-1 or type P-19 preservative. The preservatives specified and the procedures of application shall be in accordance with MIL-P-116.

5.2.1.3 Unit packs. Unless otherwise specified (see 6.2.1), each frame straightening unit shall be individually unit packed in accordance with method I of MIL-P-116. Parts preserved with type P-2 preservative shall be overwrapped with barrier material conforming to MIL-B-121, grade A. All critical surfaces shall be protected with cushioning material conforming to PPP-C-843, type II, class B. The barrier materials shall be secured with tape conforming to PPP-T-60. Small components shall be packaged in one or more fiberboard containers conforming to PPP-B-636, class weather-resistant and secured to the basic unit. When an open crate is utilized as the shipping container, the packaged components shall be packed in one or more nailed wood boxes conforming to PPP-B-621, class 1 and secured to the base of the crate.

5.2.2 Industrial preservation. When industrial preservation is specified (see 6.2.1), it shall conform to the requirements of ASTM D 3951-82.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.

5.3 Packing. Packing shall be level A or industrial as specified (see 6.2.1).

5.3.1 Level A. Each frame straightening unit shall be packed in one or more crates conforming to MIL-C-104 or MIL-C-3774. Blocking, bracing and anchoring of the items shall be in accordance with MIL-STD-1186. When MIL-C-3774 crates are utilized, all unpainted surfaces of this item shall be covered with a shroud. Shrouding requirements shall be in accordance with MIL-STD-1186.

5.3.2 Industrial packing. When industrial packing is specified (see 6.2.1), it shall conform to the requirements of ASTM D 3951-82.

5.4 Marking. In addition to any special markings specified in the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 for military marking or ASTM D 3951-82 for industrial marking (see 6.2.1).

6 NOTES

6.1 Intended use. This specification covers a motor vehicle frame straightening unit which consists principally of a rack, frame straightening attachments, force exerting units, and axle, frame and wheel alignment gages. The frame straightening unit is essentially utilized to check, diagnose, locate, and correct chassis misalignments of vehicles without removal of the frame or axle from the vehicle. The frame straightening unit is used to locate and correct area of frame misalignment due to sag, sway, mash, buckle, and diamond. The frame straightening unit is used to locate and correct, without heating, "I" beam front axle and tubular front and rear axle housing misalignments due to buckling, sag, twist, and incorrect camber. The frame straightening unit includes gages for checking vehicle wheel alignment characteristics involving caster, camber, steering axis inclination, toe-in, and turning radius. The frame straightening unit is utilized mainly in base shops and its installation is of a permanent nature.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article, when required (see 3.1).

c. Special safety devices, when required (see 3.4.4).

d. Painting, if different (see 3.7).

e. First article inspection, if different (see 4.2.1).

f. Quality conformance inspection, if different (see 4.2.2).

g. Level of preservation, packing, and marking required (see 5.2, 5.3 and 5.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.

6.2.2 Data requirements. Required technical data such as operator's manual, parts list, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes, with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - AL

Review Activities:
Army -

User Activities:
Army -

Preparing Activity:
Army - AL

Project No. 4910-A463

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament Research and
Development Center
Attn: SMCAR-ETS (R)
Rock Island, IL 61299-7300

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-45016C	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:14Z
Check the source to verify that this is the current version before use.